8	**(a)**	**(i)**	probability of decay (of a nucleus)	M1	
			<u>per</u> unit time	A1	[2]
		(ii)	$\lambda t_{½} = \ln 2$		
			$\lambda = \ln 2/(3.82 \times 24 \times 3600)$	M1	
			$= 2.1 \times 10^{-6}\ \text{s}^{-1}$	A0	[1]
	(b)		$A = \lambda N$	C1	
			$200 = 2.1 \times 10^{-6} \times N$	C1	
			$N = 9.5 \times 10^{7}$		
			ratio $= (2.5 \times 10^{25})/(9.5 \times 10^{7})$		
			$= 2.6 \times 10^{17}$	A1	[3]

Section B

9 (a) any value greater than, or equal to, $5\,\text{k}\Omega$ B1 [1]